MODERN PCB DESIGN PRACTICES

Skill-Based Course | Project Work (100 Marks)

Design of a Compact Double-Sided PCB for Automatic Room Light Controller with Visitor Counter

using Modern Design Practices

PROJECT DOCUMENTATION REPORT

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Declaration

I, Harsh Parmar, hereby declare that the project report entitled “Design of a Compact Double-Sided PCB for Automatic Room Light Controller with Visitor Counter using Modern Design Practices” is the outcome of my own individual work carried out under the guidance of the course faculty.

The schematic, PCB layout, routing, design-rule checks, 3D visualisation and fabrication outputs presented here were created by me using KiCad. Wherever the work of others, datasheets or published documentation has been consulted, it has been duly acknowledged in the References section. This work has not been copied from any other source and has not been submitted for any other course.

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I also acknowledge the open-source KiCad community, whose documentation and tools made professional-grade PCB design accessible, and the authors of the ESP32, L7805 and SSD1306 datasheets that were referred to during component selection.

Finally, I thank my family for their support and motivation throughout the course.

Harsh Parmar

Abstract

This project presents the design of a compact double-sided printed circuit board (PCB) for an Automatic Room Light Controller with Visitor Counter. Two infrared (IR) proximity sensors placed in sequence detect the direction of movement through a doorway; an ESP32 microcontroller interprets the order of sensor triggering to increment or decrement the visitor count, displays the live count on an OLED screen, and drives a relay through an NPN transistor stage so that the room light is switched on automatically when at least one person is present and off when the room is empty.

The 121 mm × 66 mm board is powered through a 12 V DC barrel jack or a USB-C connector, with an L7805 linear regulator and decoupling capacitors providing a stable 5 V rail. Modern PCB design practices were applied throughout: section-wise component placement, a continuous ground plane on the bottom copper layer, thermal-relief pad connections, current-based trace-width selection, ground-via stitching to remove isolated copper islands, via optimisation, and signal-integrity-aware routing. The layout was verified with the Design Rule Check, achieving zero errors and zero unconnected items, and was exported as Gerber and Excellon drill files, ready for fabrication.

Keywords: *Double-sided PCB, KiCad, ESP32, IR sensor, visitor counter, ground plane, via stitching, DRC, Gerber, manufacturability.*

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Chapter 1: Introduction

1.1 Background

Automatic control of lighting is one of the simplest and most effective ways to reduce wasted electrical energy in classrooms, offices, laboratories and homes. A light that stays on in an empty room is a common source of avoidable consumption. A visitor-counting controller solves this by keeping track of how many people are inside and switching the load only when the count is greater than zero.

Such a circuit can be built on a breadboard or perfboard, but a properly designed PCB is what turns a prototype into a reliable product. A PCB gives repeatable electrical behaviour, mechanical robustness, compact size and manufacturability. This project therefore focuses not only on the function of the circuit but also on how it is physically realised using modern PCB design practices.

1.2 Problem Statement

To design a compact, reliable and manufacturable double-sided PCB for an automatic room light controller that counts visitors entering and leaving through a doorway using two IR sensors, displays the count on an OLED display and switches a light through a relay, while following industry-standard PCB design rules and generating complete fabrication-ready output files.

1.3 Objectives

- Design the complete schematic of the visitor counter and light controller in KiCad.
- Select suitable components and assign verified footprints.
- Create a compact 121 mm × 66 mm double-sided PCB layout with section-wise placement.
- Provide a solid ground plane on the bottom layer, with thermal relief and ground-via stitching, for noise immunity.
- Use trace widths well above the current-carrying requirement and route with manufacturable clearances.
- Achieve a clean Design Rule Check with zero errors and zero unconnected items.
- Generate Gerber layers, Excellon drill files, BOM and 3D views for fabrication.

1.4 Scope of the Project

The scope covers schematic capture, footprint assignment, board outline definition, placement, routing, copper-zone design, ERC/DRC verification, 3D inspection and generation of fabrication outputs. The relay isolates the switched load from the control electronics through its contacts, which are brought out on a three-pin screw terminal. Firmware development and enclosure design are outside the scope of the PCB assignment and are listed under future work.

1.5 Motivation for Modern PCB Design Practices

Older, hand-drawn or single-sided layouts often suffered from long return paths, jumper wires, poor grounding and inconsistent clearances. Modern practice treats the PCB as an engineered component of the electronic system. The key motivations are:

- **Noise immunity:** a continuous ground plane gives low-impedance return paths and shields signals.
- **Reliability:** thermal relief and correct annular rings prevent soldering defects and weak joints.
- **Manufacturability:** design rules matched to fabricator capability reduce rejected boards.
- **Compactness:** two copper layers and optimised vias allow dense, tidy routing.
- **Verification:** ERC and DRC catch errors before money is spent on fabrication.

Chapter 2: System Overview and Block Diagram

2.1 System Description

The Automatic Room Light Controller with Visitor Counter monitors a doorway with two IR proximity sensors, Sensor_A and Sensor_B, mounted a short distance apart along the direction of passage. The order in which they are interrupted indicates whether a person is entering or leaving. An ESP32 development board processes these events, maintains the count, updates the OLED display and controls the light relay.

2.2 Block Diagram

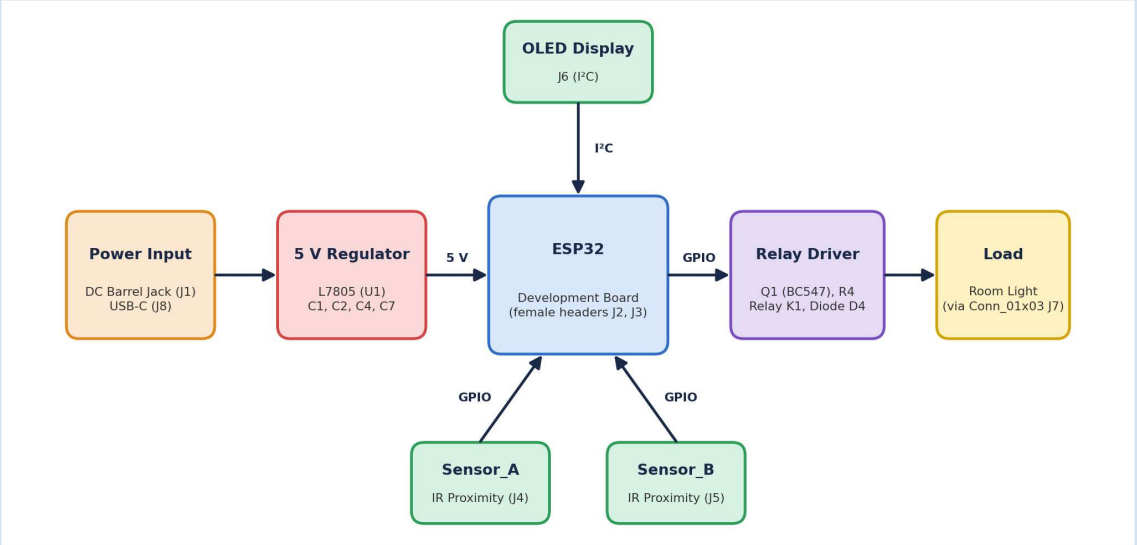


Figure 2.1: System block diagram of the visitor counter and automatic light controller

The system consists of the following functional blocks:

- **Power input:** 12 V DC barrel jack (J1) with slide switch SW2, or the power-only USB-C connector (J8).
- **5 V regulator:** L7805 (U1) with filter capacitors C1–C4 generates the 5 V rail; D4 (1N5819) lets USB VBUS feed the same rail.
- **Processing:** ESP32 development board plugged into the two 1×15 female headers J2 (J_LEFT) and J3 (J_RIGHT).
- **Sensing:** two IR proximity sensors on 3-pin headers J4 (Sensor_A) and J5 (Sensor_B), with 10 kΩ pull-up resistors R2 and R3.
- **Display:** I²C OLED display on 4-pin header J6.
- **Load control:** BC547 transistor Q1 with base resistor R4 drives relay K1; D3 (1N4007) is the flyback diode; the relay contacts go to the 3-pin screw terminal J7.

2.3 Working Principle

2.3.1 Direction Detection

When a person enters, they interrupt Sensor_A first and Sensor_B second; when leaving, the order is reversed. The firmware records which sensor triggered first and increments the counter for the sequence A→B (entry) or decrements it for B→A (exit). A short debounce delay and a timeout between the two events prevent false counts.

2.3.2 Counting and Display

The ESP32 holds the current occupancy count in memory and never lets it fall below zero. After every valid event the count is refreshed on the OLED over the I^2C interface so that occupants and maintainers can see the room status at a glance.

2.3.3 Automatic Light Control

If the count is greater than zero, the ESP32 sets the control GPIO high. This saturates transistor Q1 through base resistor R4, energising the coil of relay K1 and closing its contacts so the connected light turns on. When the count returns to zero, the GPIO goes low, Q1 turns off and the relay releases. Diode D3 across the coil absorbs the inductive voltage spike when the coil is de-energised.

2.3.4 Operating Flow

1. Power is applied through J1 (with SW2 on) or J8; the 5 V rail stabilises and the ESP32 boots.
2. The count is initialised to zero and the relay is held off.
3. Sensor events are read continuously; direction is resolved from the order of triggering.
4. The count is updated and shown on the OLED.
5. The relay is energised if the count is above zero, otherwise it is released.

Chapter 3: Hardware Components and Bill of Materials

3.1 Bill of Materials

The Bill of Materials (BOM) below lists every component on the board with its reference designator, value, quantity and footprint, exactly as assigned in KiCad's Footprint Assignment tool (see Figure 4.5). The BOM is also exported from KiCad as a CSV file and submitted with the source files.

Table 3.1: Bill of Materials

Ref.	Component	Value / Spec.	Qty	Footprint (KiCad library)	Package
U1	Linear voltage regulator	L7805 (5 V, 1 A)	1	Package_TO_SOT_THT:TO-220-3_Vertical	TO-220
—	ESP32 development board (30-pin)	ESP32 DevKit	1	Plugs into J2 and J3	Module
J2, J3	Female headers for ESP32 (J_LEFT, J_RIGHT)	1×15, 2.54 mm	2	Connector_PinHeader_2.54mm:PinHeader_1x15_P2.54mm_Vertical	THT
J1	DC barrel jack	12 V input	1	Connector_BarrelJack:BarrelJack_Horizontal	THT
J8	USB-C receptacle, power-only 6-pin	USB Type-C	1	Connector_USB:USB_C_Receptacle_HRO_TYPE-C-31-M-12	SMD
SW2	Slide switch SPDT (power switch)	SPDT	1	Button_Switch_THT:SW_Slide_SPDT_Straight_CK_OS102011MS2Q	THT
C1	Electrolytic capacitor (input bulk)	470 µF	1	Capacitor_THT:CP_Radial_Tantal_D8.0mm_P2.50mm	Radial
C3	Electrolytic capacitor (output bulk)	100 µF	1	Capacitor_THT:CP_Radial_Tantal_D8.0mm_P2.50mm	Radial
C2, C4	Ceramic bypass capacitor	100 nF	2	Capacitor_THT:C_Disc_D3.8mm_W2.6mm_P2.50mm	Disc
D2, D4	Schottky diode	1N5819	2	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal	DO-41
D3	Flyback diode (relay coil)	1N4007	1	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal	DO-41
D1	Power indicator LED	5 mm, red	1	LED_THT:LED_D5.0mm	5 mm
R1, R4	Resistor (LED / base current limit)	1 kΩ	2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Axial
R2, R3	Pull-up resistor (sensor inputs)	10 kΩ	2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Axial
R5, R6	USB-C CC pull-down resistor	5.1 kΩ	2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Axial

Q1	NPN switching transistor	BC547	1	Package_TO_SOT_THT:TO-92_Inline	TO-92
K1	Relay, SPDT (Form C), 5 V coil	SANYOU SRD series	1	Relay_THT:Relay_SPDT_SANYOU_SRD_Series_Form_C	THT
J7	Screw terminal, 3-pin (relay contacts / load)	5.08 mm pitch	1	TerminalBlock_Phoenix:TerminalBlock_Phoenix_MKDS-1,5-3-5.08_1x03_P5.08mm_Horizontal	THT
J4, J5	IR sensor headers (Sensor_A, Sensor_B)	1×03, 2.54 mm	2	Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical	THT
J6	OLED I²C header	1×04, 2.54 mm	1	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical	THT

3.2 Key Components

- **ESP32 development board:** dual-core 32-bit MCU with Wi-Fi and Bluetooth, many GPIOs and I²C support; 3.3 V logic; sockets into the board so it can be reused.
- **IR proximity sensors:** low-cost, contactless detection of a person crossing the beam, with a simple digital output.
- **OLED display (I²C):** high-contrast, low-power display needing only four connections.
- **L7805 regulator:** simple, robust 5 V regulation with thermal shutdown and short-circuit protection.
- **Relay K1 with driver Q1:** the GPIO cannot drive a relay coil directly, so BC547 acts as a switch; the contacts isolate the load from the logic.
- **USB-C input (J8):** power-only connector; 5.1 kΩ resistors R5 and R6 on CC1/CC2 let a USB-C source supply 5 V.

3.3 Component Selection Justification

Table 3.2: Component selection justification

Component	Reason for selection
ESP32 DevKit	Sufficient GPIOs, I²C support, 3.3 V logic and room for IoT or app expansion; widely available.
L7805 (U1)	Converts the 12 V adapter input to a stable 5 V for relay, sensors, OLED and ESP32.
470 µF / 100 µF + 100 nF capacitors	Bulk capacitors handle load transients; 100 nF ceramics bypass high-frequency noise at the regulator.
BC547 (Q1) with R4 = 1 kΩ	Handles the relay coil current (about 70–90 mA) comfortably; low cost, TO-92 package.
1N4007 (D3)	Clamps the relay-coil back-EMF; 1 A rating gives ample margin.
1N5819 (D2, D4)	Schottky diodes have a low forward drop, so little voltage is lost on the 5 V path.

SW2 and LED D1	SW2 provides a manual power switch; D1 with R1 gives a power-on indication.
Through-hole parts	Easy hand soldering and inspection during prototyping and viva demonstration.

Chapter 4: Schematic Design

4.1 Schematic Philosophy

The schematic (project ESP32_Visitor_Counter, A3 sheet, revision 1.0) was drawn in KiCad's Schematic Editor with functional blocks arranged logically: power input and regulation, ESP32 headers with sensors and OLED, and the relay output stage. Power rails (+12 V, +5 V, +3.3 V and GND) use power symbols, and PWR_FLAG symbols mark the driven power nets so the checker recognises them.

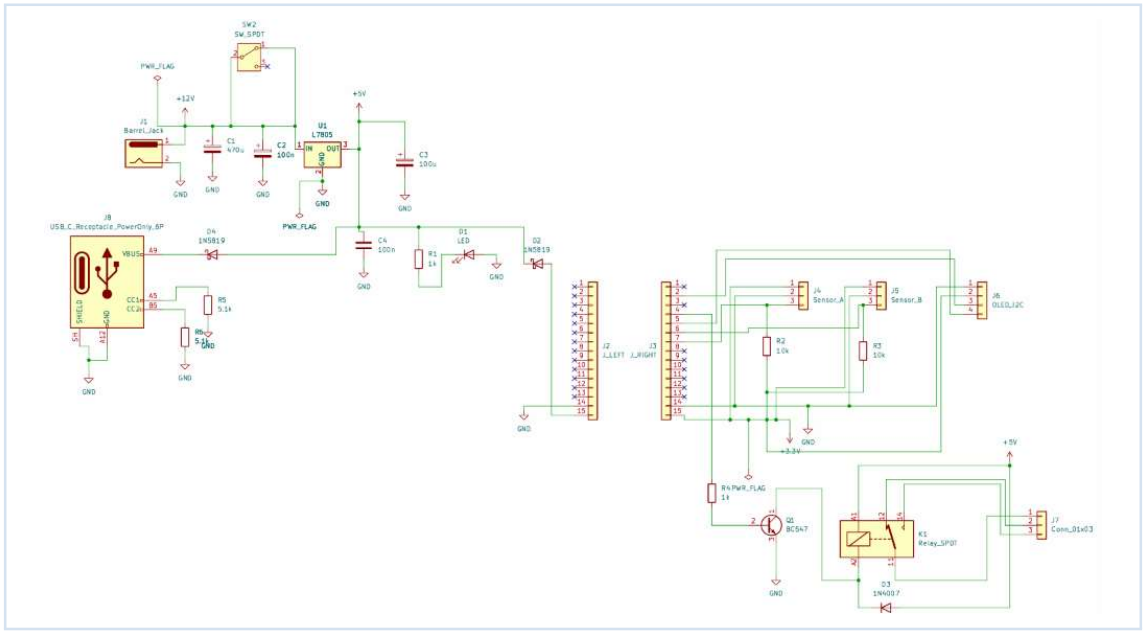


Figure 4.1: Complete schematic diagram of the visitor counter board

4.2 Power Supply Section

Power can be applied through the barrel jack J1 or the USB-C connector J8. On the 12 V path, slide switch SW2 connects the input to C1 (470 µF bulk) and C2 (100 nF bypass) ahead of the L7805 regulator U1. The 5 V output is smoothed by C3 (100 µF) and C4 (100 nF). Schottky diode D4 connects USB VBUS (pin A9) to the 5 V rail, and resistors R5 and R6 (5.1 kΩ) on CC1 and CC2 identify the board as a USB-C power sink. LED D1 with R1 indicates that the rail is live, and D2 is a further 1N5819 Schottky diode on the 5 V branch.

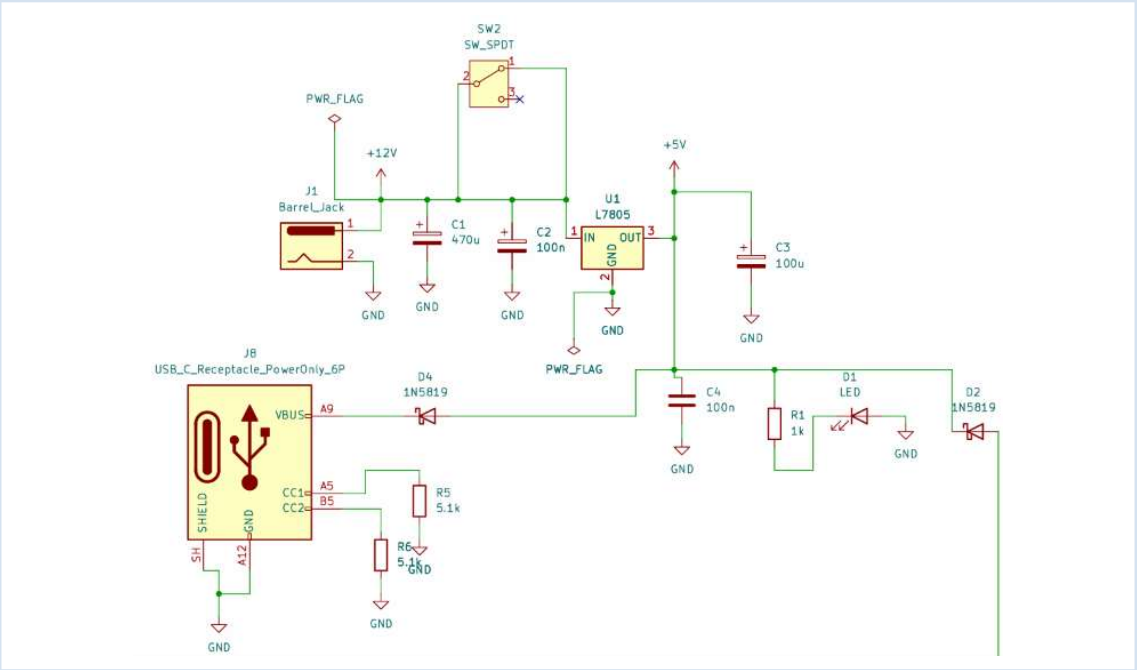


Figure 4.2: Power supply section – J1, J8, SW2, U1 (L7805), C1–C4, D4, R5, R6

4.3 Sensor and Display Interfacing

Sensor_A (J4) and Sensor_B (J5) are three-pin headers (VCC, GND, OUT), with their OUT lines routed to ESP32 GPIO inputs on J2/J3 and held at a defined level by 10 kΩ resistors R2 and R3. The I²C OLED connects through header J6 (SDA, SCL, power and ground). Keeping the sensor and OLED headers near the board edge allows short wiring to the doorway sensors and the display.

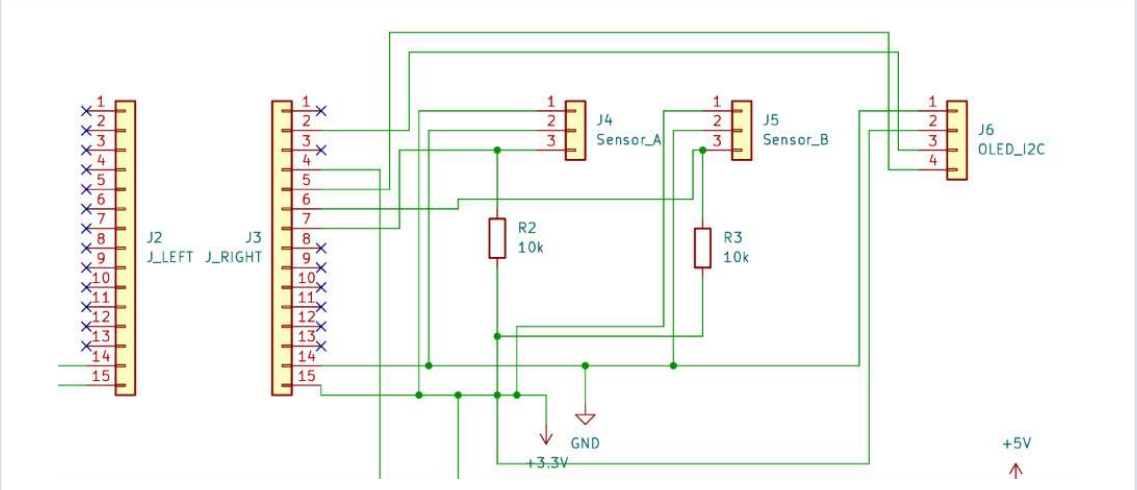


Figure 4.3: ESP32 headers J2/J3, Sensor_A (J4), Sensor_B (J5) and OLED header (J6)

4.4 Output Driving Circuit

The ESP32 control GPIO connects to the base of NPN transistor Q1 (BC547) through R4 (1 kΩ). The emitter is grounded and the collector sinks the coil current of relay K1. Flyback diode D3 (1N4007)

sits across the coil to suppress the voltage spike produced when the coil is switched off, protecting Q1 and the neighbouring circuitry. The relay contacts are brought out to the three-pin screw terminal J7.

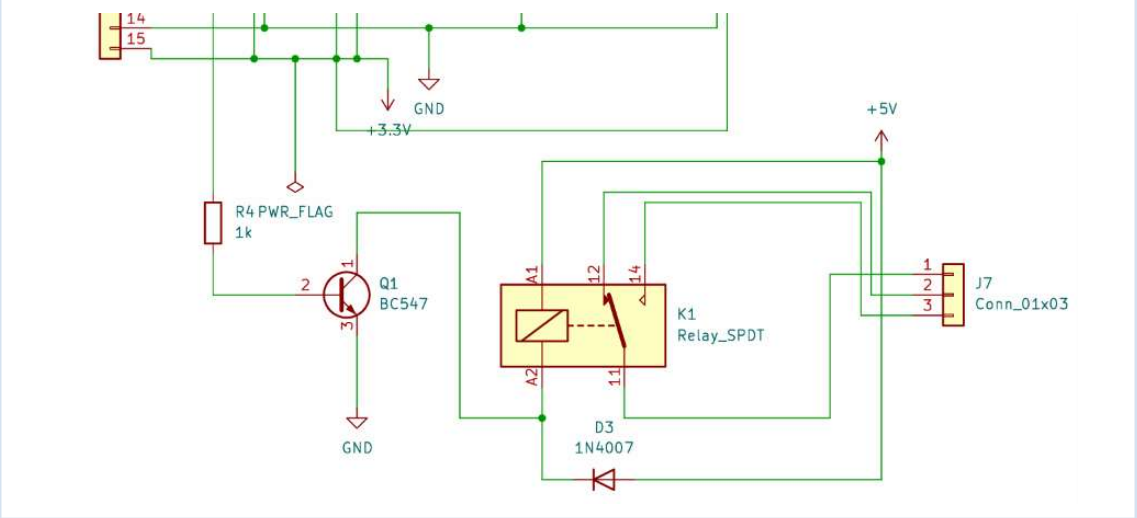


Figure 4.4: Relay driver section – Q1, R4, D3, K1 and load terminal J7

4.5 Footprint Assignment

Each symbol was assigned a footprint using KiCad's Footprint Assignment tool and the standard libraries, as listed in Figure 4.5. Pad sizes, drill diameters and pitches were compared with datasheet dimensions before layout began. The USB-C connector J8 uses the HRO TYPE-C-31-M-12 footprint.

Symbol : Footprint Assignments

1	C1	470u	Capacitor_THT:CP_Radial_Tantal_D8.0mm_P2.50mm
2	C2	100n	Capacitor_THT:C_Disc_D3.8mm_W2.6mm_P2.50mm
3	C3	100u	Capacitor_THT:CP_Radial_Tantal_D8.0mm_P2.50mm
4	C4	100n	Capacitor_THT:C_Disc_D3.8mm_W2.6mm_P2.50mm
5	D1	LED	LED_THT:LED_D5.0mm
6	D2	1N5819	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal
7	D3	1N4007	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal
8	D4	1N5819	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal
9	J1	Barrel_Jack	Connector_BarrelJack:BarrelJack_Horizontal
10	J2	J_LEFT	Connector_PinHeader_2.54mm:PinHeader_1x15_P2.54mm_Vertical
11	J3	J_RIGHT	Connector_PinHeader_2.54mm:PinHeader_1x15_P2.54mm_Vertical
12	J4	Sensor_A	Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical
13	J5	Sensor_B	Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical
14	J6	OLED_I2C	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical
15	J7	Conn_01x03	TerminalBlock_Phoenix:TerminalBlock_Phoenix_MKDS-1,5-3-5.08_1x03_P5.08mm_Horizontal
16	J8	USB_C_Receptacle_PowerOnly_6P	Connector_USB:USB_C_Receptacle_HRO_TYPE-C-31-M-12
17	K1	Relay_SPDT	Relay_THT:Relay_SPDT_SANYOU_SRD_Series_Form_C
18	Q1	BC547	Package_TO_SOT_THT:TO-92_Inline
19	R1	1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
20	R2	10k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
21	R3	10k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
22	R4	1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
23	R5	5.1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
24	R6	5.1k	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
25	SW2	SW_SPDT	Button_Switch_THT:SW_Slide_SPDT_Straight_CK_OS102011MS2Q
26	U1	L7805	Package_TO_SOT_THT:TO-220-3_Vertical

Figure 4.5: Component selection and footprint assignment (26 symbols)

4.6 Electrical Rules Check

The Electrical Rules Check (ERC) was run on the finished schematic to confirm that no pins are left unconnected, no power pins are undriven and no nets are shorted. The only message is a library

warning that the Relay_SPDT symbol differs from its library copy; this concerns the symbol graphics only and does not affect the netlist.

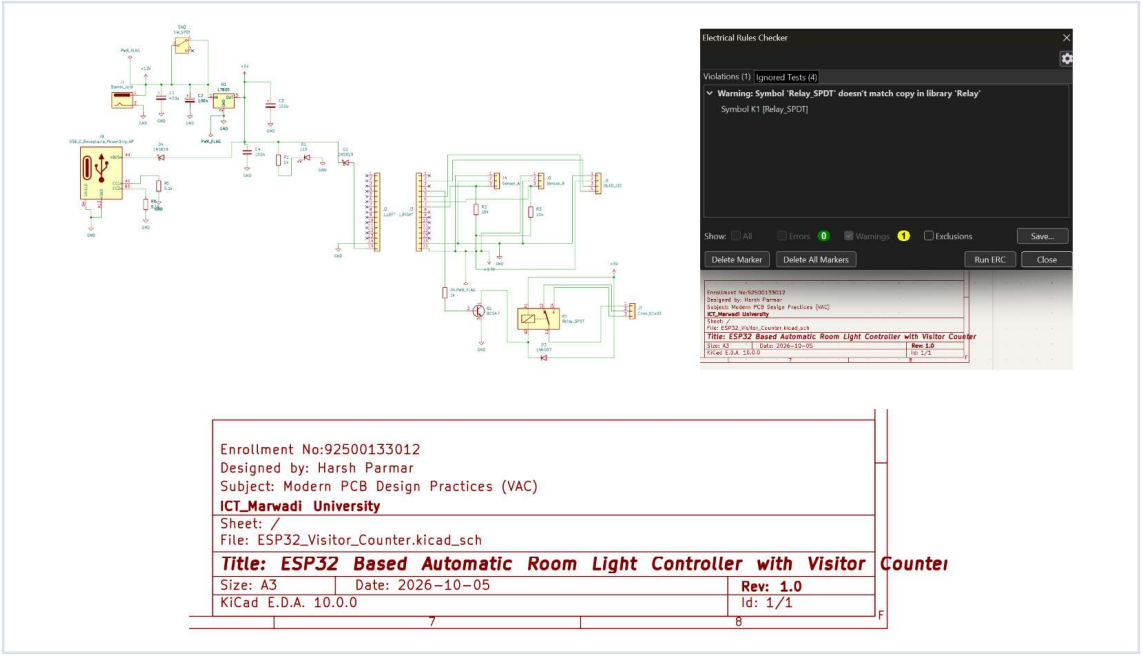


Figure 4.6: Electrical Rules Check result and schematic title block

Chapter 5: PCB Layout Design Using Modern Practices

5.1 Why a Double-Sided PCB

A double-sided board provides two copper layers: the top layer carries all signal routing, while the bottom layer is dedicated to an unbroken ground plane. Compared with a single-sided design it eliminates jumper wires, shortens return paths, reduces loop area and allows a smaller board. For a design combining an ESP32, an I²C display, a relay and a linear regulator, this gives clearly better noise performance and reliability.

5.2 Board Outline and Component Placement

The board outline was defined on the Edge.Cuts layer as 121 mm × 66 mm. Components were placed section-wise to keep functions separate and traces short:

- **Power section:** J1, SW2, U1 and the filter capacitors on the right side, together with J8, so high-current paths and regulator heat stay away from the logic.
- **ESP32 section:** the two 1×15 headers J2 and J3 occupy the left and centre of the board, giving short routes to all peripherals.
- **Relay section:** K1, Q1, R4, D3 and the screw terminal J7 are grouped at the top, away from the sensor lines.
- **Sensors and OLED:** J4, J5 and J6 are placed along the board edges for easy cable access.

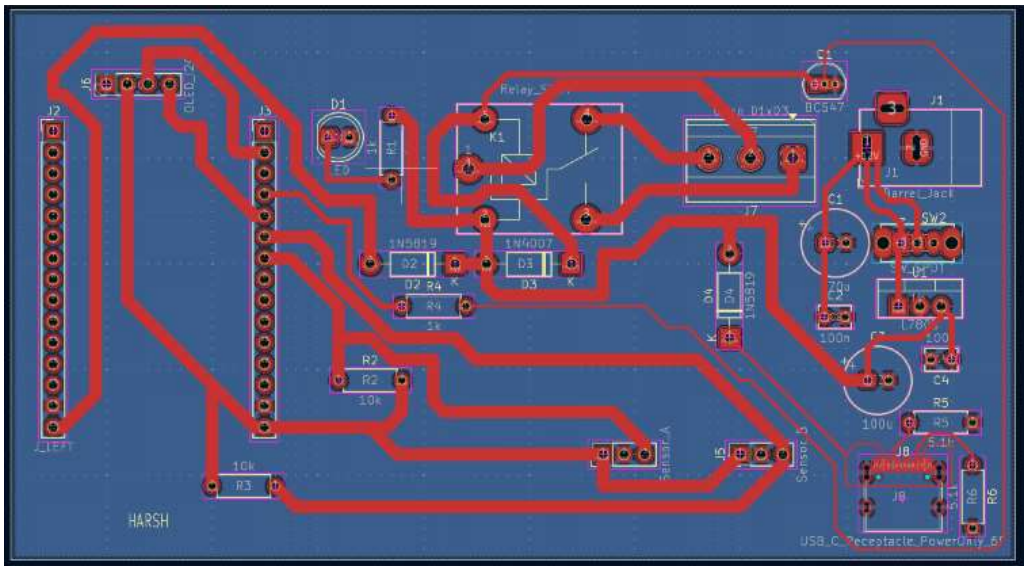


Figure 5.1: Board outline (121 mm × 66 mm), component placement and top copper layer (F.Cu)

5.3 Design Parameters

Table 5.1: Board and design parameters

Parameter	Value	Remark

Board size	121 mm × 66 mm	Compact form factor
Layers	2 (F.Cu, B.Cu)	Double-sided, 1.6 mm FR-4
Copper weight	1 oz (35 µm)	Standard
Trace width	≥ 1.0 mm	Wide routing on all nets, far above the current requirement
Minimum clearance	0.2 mm	Within standard fabricator capability
Via size / drill	0.8 mm / 0.4 mm	Standard, low-cost drill
Ground	Solid GND zone on B.Cu	Top-side ground pads tied to it with GND vias

5.4 Ground Plane Design

A filled GND copper zone covers the whole bottom layer (B.Cu, Figure 6.3). Ground pads on the top layer connect to it through vias and thermal-relief connections. A continuous ground plane gives signals a nearby low-impedance return path, reduces loop area and electromagnetic interference, and acts as a heat spreader around the regulator. The zone clearance was matched to the design rules so the fill never violates spacing to other nets.

5.5 Thermal Relief

Through-hole pads connected to the ground pour use thermal-relief spokes rather than solid connections. The spokes limit heat conduction away from the pad during soldering, so the joint reaches soldering temperature and cold joints are avoided, while still giving a low-resistance electrical path in operation.

Regulator U1 deserves special attention. With a 12 V input and a load of roughly 0.25–0.3 A, the L7805 dissipates (12 V − 5 V) × I ≈ 1.8–2.1 W. A TO-220 package without a heat sink will run hot at this level, so U1 is surrounded by copper and a small clip-on heat sink is recommended. When the board is powered from USB-C the regulator is bypassed and dissipates nothing.

5.6 Trace Width Calculation

Trace widths were checked with the IPC-2221 relationship for external layers, $I = k \cdot \Delta T^{0.44} \cdot A^{0.725}$, where $k = 0.048$, ΔT is the temperature rise in °C and A is the cross-sectional area in mil². For 1 oz copper (about 1.38 mil thick) and a 10 °C rise, the minimum widths are:

Table 5.2: Trace width check (IPC-2221, external layer, 1 oz copper, 10 °C rise)

Net type	Typical current	Min. calculated width	Width used
Sensor / I²C / GPIO signals	< 20 mA	< 0.1 mm	≥ 1.0 mm
Relay coil path (Q1, D3)	≈ 80 mA	≈ 0.1 mm	≥ 1.0 mm
5 V distribution	≈ 0.3 – 0.5 A	≈ 0.15 – 0.2 mm	≥ 1.0 mm
Relay contact / load path	up to 1 A	≈ 0.3 mm	≥ 1.0 mm

All nets are routed well above the calculated minimum. The generous width lowers voltage drop and temperature rise, makes the board tolerant of etching variation and is easy to manufacture.

5.7 Routing Strategy

Routing was carried out interactively with 45° bends and short, direct paths. Power and relay-driver nets were routed first, followed by the sensor, display and ESP32 signal nets. Right-angle corners were avoided, and the relay-coil loop (K1, Q1, D3) was kept tight and apart from the sensor lines. All signal tracks were routed on the top layer (Figures 5.1 and 6.2), which leaves the bottom layer free for an unbroken ground plane (Figure 6.3).

5.8 Via Optimisation and Ground Stitching

Vias were kept to a standard 0.8 mm / 0.4 mm size so they stay inexpensive and reliable. After the first routing pass, 21 unconnected items were reported in the ratsnest (Figure 5.2). Analysis showed that the signal traces had divided the GND copper into small isolated islands, so ground could not reach every pad.

The problem was solved with GND via stitching. Ground vias were placed (V key) near the unconnected pads around C1, C2, the ESP32 headers, the sensors and the OLED, linking the top-side ground pads to the bottom ground plane. After refilling all zones (B key) the unconnected count fell from 21 to 0. Stitching also reduces ground impedance, improves return-path continuity and strengthens shielding.

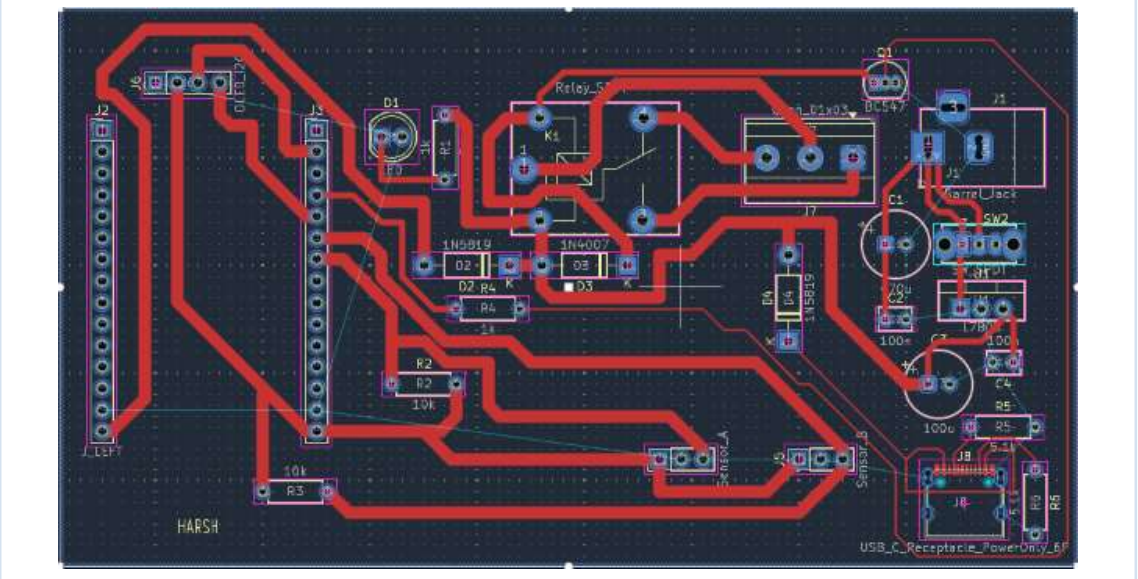


Figure 5.2: Routed board with unconnected (ratsnest) items before GND via stitching and zone refill

5.9 Signal Integrity Considerations

- Short, direct signal traces keep inductance and coupling low.

- The continuous ground plane gives every signal an adjacent return path.
- The I²C lines to the OLED are routed together and kept away from the relay and regulator section.
- Bypass capacitors C2 and C4 sit close to the regulator pins.
- The relay coil and its flyback diode D3 form a tight loop, away from the sensor lines.

5.10 Issues Faced and Resolutions

Table 5.3: Issues faced and resolutions

Issue	Cause	Resolution
"No net" errors on J8 (USB-C)	Dual-pad structure of the connector footprint left overlapping pads without a net.	Assigned unified nets: A12 to GND and A9 to VBUS.
NPTH hole clearance violation	Interactive router passed a trace over a mounting hole of the USB-C connector.	Changed router mode to "Walk around" and re-routed along a safe path.
21 unconnected items	Signal traces split the GND pour into isolated islands.	Placed GND stitching vias and refilled zones: result 0 unconnected.
Library mismatch warning (J8)	Footprint was edited locally and differs from the library copy.	Verified the geometry manually; the warning does not affect manufacturing.
Remaining signal connections	R4–Q1 base and the last diode pins were still open.	Routed manually and re-ran DRC.

5.11 Design Rule Check (DRC)

The Design Rules Check covers clearance, track width, via size, annular ring, hole-to-hole spacing, board-edge clearance, silkscreen overlap and unconnected items. The final result is shown in Figure 5.3.

Check	Result
Errors	0
Unconnected items	0
Warnings	1 – library footprint mismatch on J8, safe for manufacturing

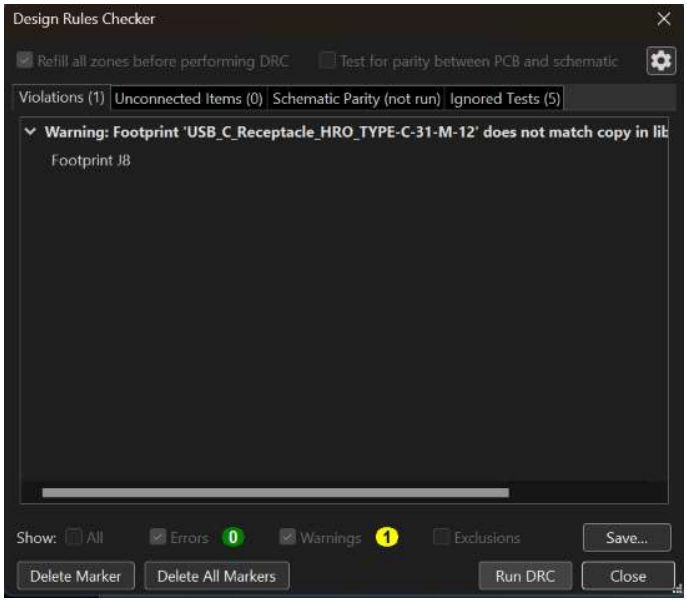


Figure 5.3: Design Rule Check result – 0 errors, 0 unconnected items, 1 library warning

5.12 Manufacturing Considerations

- **Clearance:** 0.2 mm minimum copper clearance, within typical low-cost fabricator limits.
- **Annular ring and drill:** pads and vias are sized to keep an adequate annular ring after drilling tolerance.
- **Solder mask:** mask openings follow the pad shapes with a small expansion.
- **Silkscreen:** reference designators and polarity marks are kept clear of pads; the name "HARSH" is added on F.Silkscreen.
- **Board edge:** copper and components are kept clear of the Edge.Cuts outline.

5.13 3D PCB View

The KiCad 3D Viewer was used to inspect component placement, connector orientation, board outline and silkscreen, and to confirm that no parts collide or overhang the board edge.

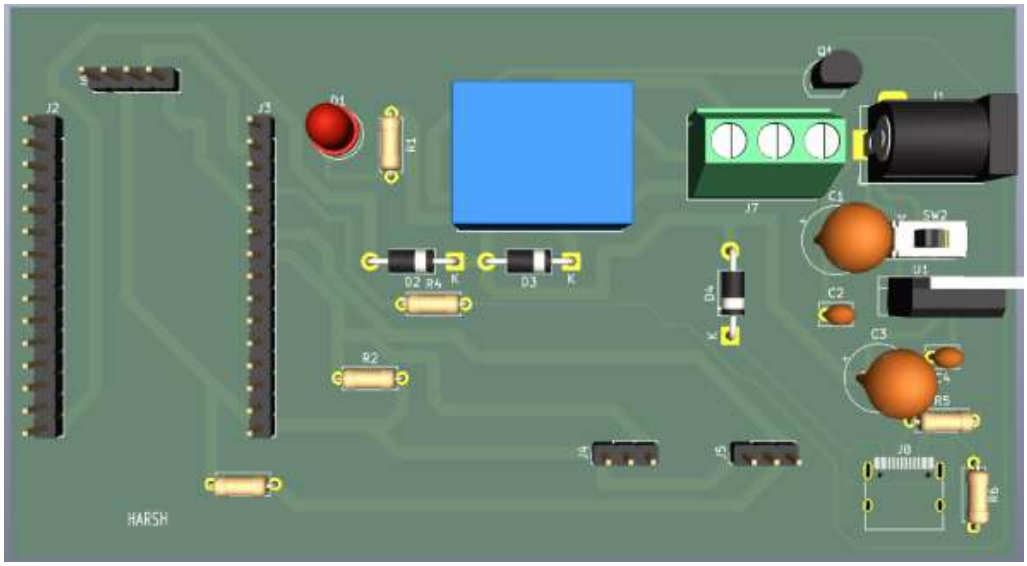


Figure 5.4: 3D PCB view – top side

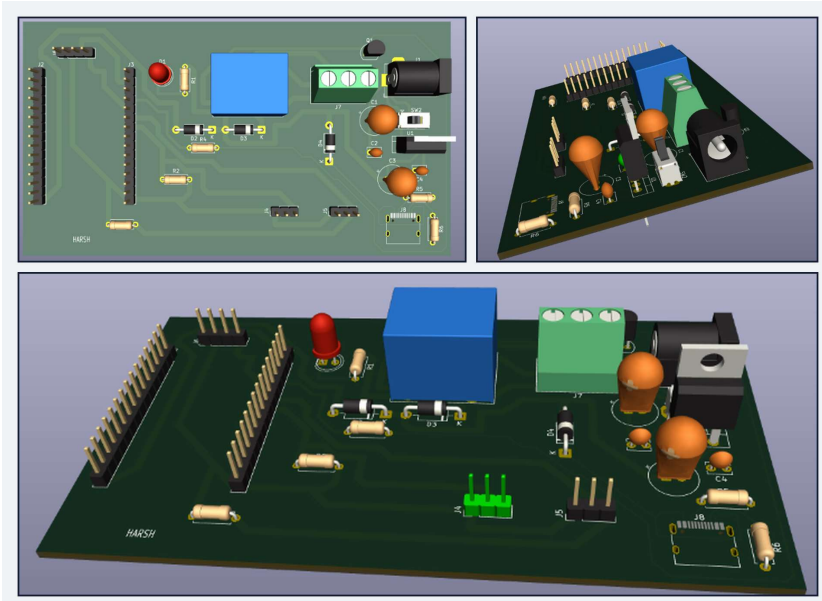


Figure 5.5: 3D PCB views from different angles

Chapter 6: Fabrication Files and Gerber Generation

6.1 Output Generation

The fabrication package was produced from KiCad's Plot dialog (Gerber, RS-274X) and the Generate Drill Files dialog (Excellon). All files carry the project prefix ESP32_Visitor_Counter. The generated files are shown in Figure 6.1, and each layer was then opened in the KiCad Gerber Viewer to confirm that the output is correct (Figures 6.2 to 6.7).

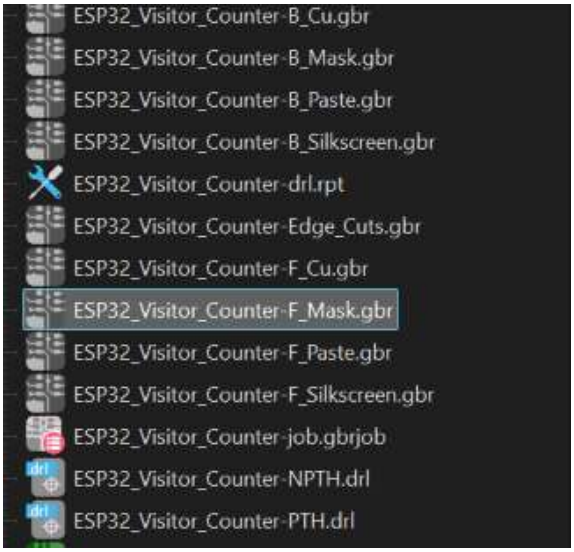


Figure 6.1: Generated Gerber and drill files in the output folder

6.2 List of Generated Files

Table 6.1: Generated fabrication files

File	Purpose
ESP32_Visitor_Counter-F_Cu.gbr	Top copper layer: pads and all signal tracks
ESP32_Visitor_Counter-B_Cu.gbr	Bottom copper layer: solid ground plane with thermal-relief pads
ESP32_Visitor_Counter-F_Mask.gbr	Top solder-mask openings
ESP32_Visitor_Counter-B_Mask.gbr	Bottom solder-mask openings
ESP32_Visitor_Counter-F_Silkscreen.gbr	Top legend: reference designators, outlines, name “HARSH”
ESP32_Visitor_Counter-B_Silkscreen.gbr	Bottom legend
ESP32_Visitor_Counter-F_Paste.gbr	Top solder-paste openings (SMD USB-C connector J8)
ESP32_Visitor_Counter-B_Paste.gbr	Bottom solder-paste layer
ESP32_Visitor_Counter-Edge_Cuts.gbr	Board outline, 121 mm × 66 mm
ESP32_Visitor_Counter-PTH.drl	Plated through-hole drill data (Excellon)

ESP32_Visitor_Counter-NPTH.drl	Non-plated hole drill data (Excellon), e.g. USB-C connector pegs
ESP32_Visitor_Counter-drl.rpt	Drill report: tool sizes and hole counts
ESP32_Visitor_Counter-job.gbrjob	Gerber job file describing the board and its layers

6.3 Explanation of Layers

Copper layers carry the electrical connections. Solder-mask layers define where the coating is opened to expose copper for soldering. Silkscreen layers carry the printed text and symbols that guide assembly. Paste layers define stencil openings for SMD pads; on this board only the USB-C connector J8 is surface-mounted, and every other part is through-hole. The Edge.Cuts layer tells the fabricator how to route the board profile, and the drill files define the position and diameter of every hole, separated into plated and non-plated types.

6.4 Gerber Verification

Each Gerber layer was inspected in the KiCad Gerber Viewer. The top copper layer carries all signal routing (Figure 6.2), while the bottom copper layer is a continuous ground plane with clearances around signal pads and thermal-relief spokes on ground pads (Figure 6.3). The outline is a closed rectangle (Figure 6.4), the silkscreen shows reference designators and the name “HARSH” (Figure 6.5), and the paste layer contains only the pads of the SMD connector J8 (Figure 6.6). Figure 6.7 shows the layers overlaid.

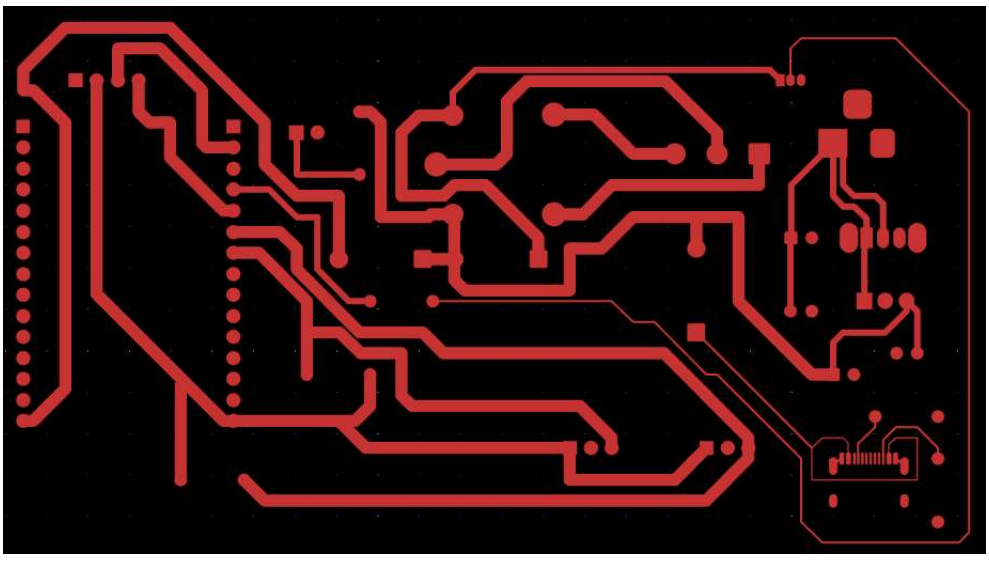


Figure 6.2: F.Cu Gerber – top copper layer with all signal tracks

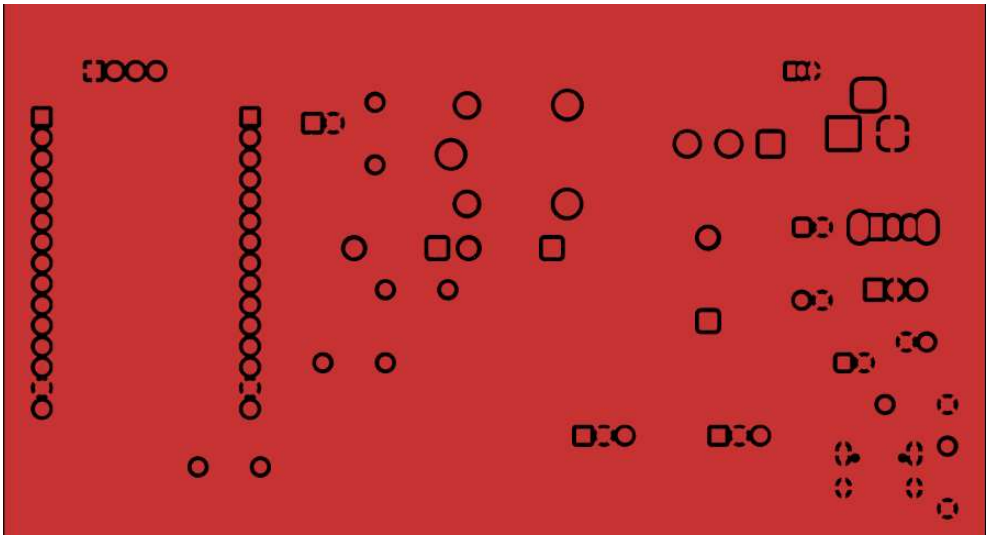


Figure 6.3: B.Cu Gerber – bottom copper layer, solid ground plane

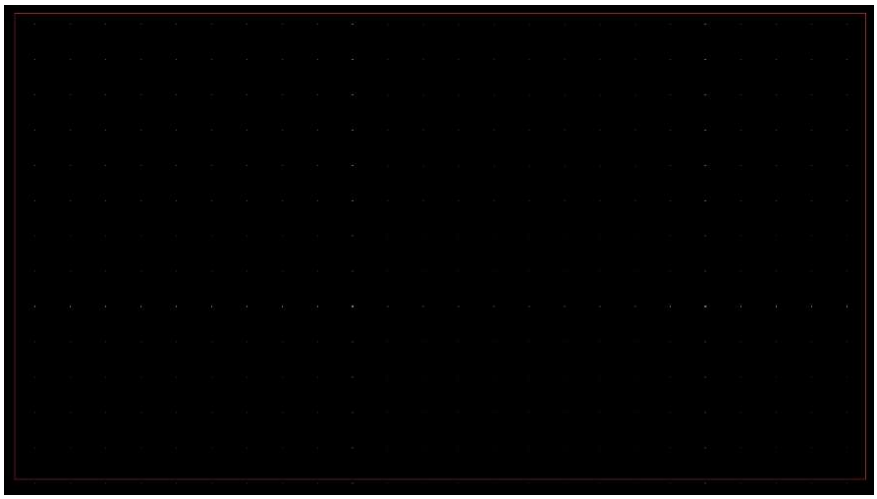


Figure 6.4: Edge.Cuts Gerber – board outline

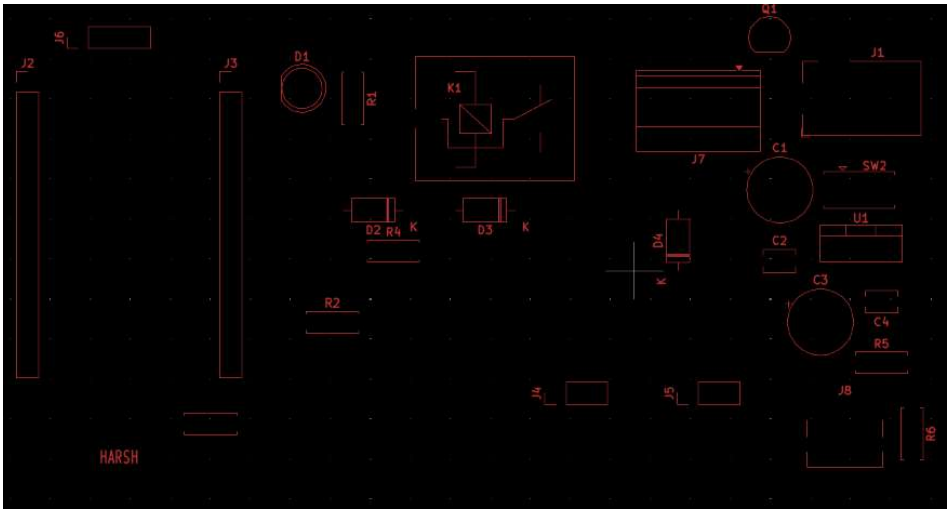


Figure 6.5: F.Silkscreen Gerber – reference designators and name "HARSH"

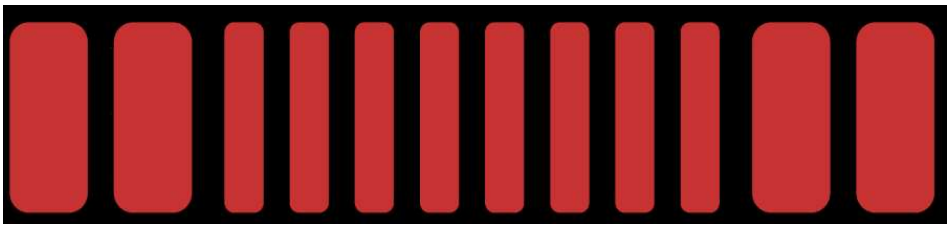


Figure 6.6: F.Paste Gerber – solder-paste openings of the SMD connector J8

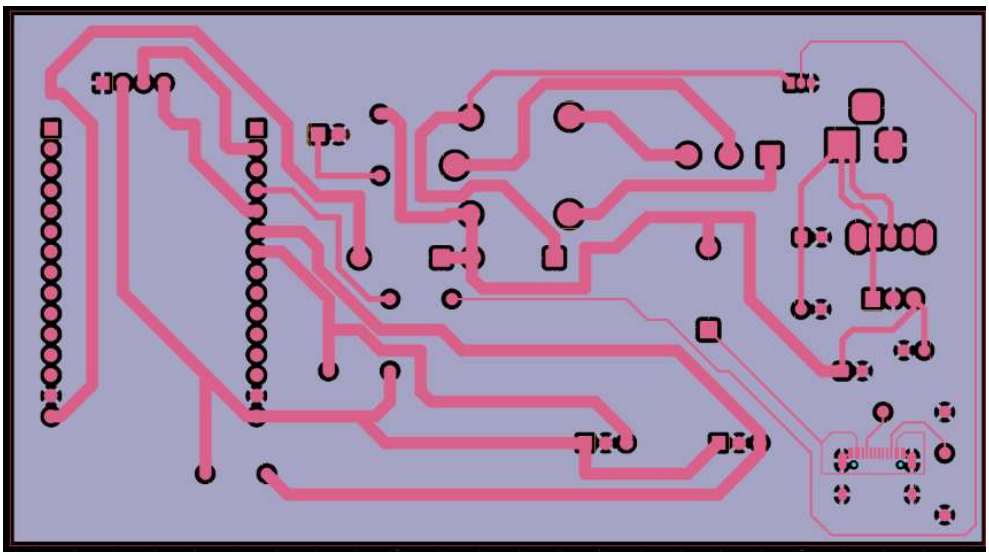


Figure 6.7: Gerber Viewer – F.Cu overlaid on the B.Cu ground plane

6.5 Readiness for Fabrication

The design is fabrication-ready because the DRC shows zero errors and zero unconnected items, all copper, mask, silkscreen, paste, outline and drill files are present, minimum clearances and sizes lie

within standard fabricator capability, and the Gerber Viewer check confirms that every layer was exported correctly. All Gerber and drill files are supplied in a ZIP folder together with the source project and BOM.

Chapter 7: Testing, Simulation and Validation

7.1 Design Verification Results

The design was verified in software at two stages. The Electrical Rules Check on the schematic (Figure 4.6) reports no connection errors, with a single library-symbol warning for the relay. The Design Rules Check on the layout (Figure 5.3) reports 0 errors and 0 unconnected items, with a single footprint-library warning for J8. The 3D views (Figures 5.4 and 5.5) confirmed component orientation and clearances.

7.2 Test Procedure for the Assembled Board

After fabrication and assembly, the following procedure is used to validate the board:

1. Visual inspection: check for solder bridges, missing parts and correct polarity of C1, C3, D1–D4 and Q1.
2. Continuity test: confirm there is no short between 5 V and GND with a multimeter before applying power.
3. Power-up test: apply 12 V (SW2 on) and measure the regulator output; expected 5 V ± 5 %. Repeat via USB-C.
4. Sensor test: block each IR sensor and verify the output level change.
5. Display test: confirm the OLED initialises and shows the count.
6. Direction test: pass an object A→B then B→A and observe the count.
7. Relay test: verify the relay clicks when the count is above 0 and releases at 0.

Table 7.1: Test cases and expected results

Test	Condition	Expected result
Power rail	12 V via J1 (SW2 on) or 5 V via J8	5 V present on the 5 V rail; LED D1 on
Entry	A triggered, then B	Count +1, light ON
Exit	B triggered, then A	Count −1, light OFF at 0
Empty room	Count = 0	Relay released, load OFF
Negative guard	Exit event at count 0	Count stays at 0
Relay back-EMF	Relay switched OFF	No stress on Q1; spike clamped by D3

7.3 Troubleshooting Guide

- **No power:** check the J1 polarity, SW2 position, C1 orientation and regulator orientation.

- **ESP32 not booting:** verify 5 V at the ESP32 VIN pin and that the board is seated correctly in J2/J3.
- **Wrong count direction:** swap the Sensor_A and Sensor_B assignment in firmware.
- **False counts:** adjust sensor sensitivity and the debounce interval.
- **Relay does not switch:** check Q1 orientation, R4 value and the 5 V coil supply.
- **Regulator hot:** add a heat sink to U1 or power the board from USB-C.

Chapter 8: Results and Discussion

8.1 Achievements

- Complete schematic and double-sided PCB designed in KiCad (project ESP32_Visitor_Counter).
- Compact 121 mm × 66 mm board with section-wise placement.
- Final DRC result: 0 errors and 0 unconnected items.
- Solid ground plane on the bottom layer, stitched to the top-side ground pads with vias; 21 initial unconnected items resolved.
- USB-C footprint and routing issues diagnosed and fixed.
- Gerber layers, Excellon drill files, BOM and 3D views generated.

8.2 Impact of Modern Practices

The bottom-layer ground plane and via stitching provide continuous return paths and reduce noise; thermal-relief connections make soldering reliable; generous trace widths avoid overheating; DRC removed layout errors before fabrication; and section-wise placement kept routing short and clean. Together these practices give a board that is more reliable and easier to manufacture than an informally drawn layout.

8.3 Comparison with Single-Sided / Older Practices

Table 8.1: Single-sided vs modern double-sided design

Aspect	Single-sided / older practice	This design (modern)
Routing layers	1, jumper wires needed	2 layers, no jumpers
Grounding	Thin ground traces	Solid ground plane on bottom layer, stitched
Noise immunity	Poor, long return paths	Good, short return paths
Soldering reliability	Solid pad-to-copper joins	Thermal-relief pads
Verification	Visual only	ERC + DRC with 0 errors
Fabrication output	Manual artwork	Gerber + Excellon, ready to send

8.4 Limitations

The L7805 is a linear regulator and dissipates heat in proportion to the input-output voltage difference, so on a 12 V input it needs a heat sink; a buck converter would be more efficient. IR sensors may be affected by strong ambient light, and the relay contact rating must be respected for the connected load. The board has been verified by ERC and DRC; functional test results will be recorded once the assembled board is tested.

Chapter 9: Conclusion and Future Scope

9.1 Conclusion

This project successfully designed a compact double-sided PCB for an Automatic Room Light Controller with Visitor Counter. The work covers the full flow from schematic capture and footprint assignment through placement, routing, ground-plane design, DRC, 3D inspection and Gerber/drill generation. Real design problems, namely the USB-C footprint nets, a router hole violation and isolated ground islands, were identified and resolved with sound engineering methods, leading to a final DRC result of zero errors and zero unconnected items.

The project demonstrates the value of modern PCB practices: ground planes, thermal relief, via stitching, generous current-based trace sizing and manufacturability checks. The resulting design is ready to be fabricated and assembled.

9.2 Future Scope

- Replace the L7805 with a buck converter for higher efficiency.
- Add a Wi-Fi dashboard or mobile application for remote occupancy monitoring using the ESP32.
- Use a solid-state relay or an optocoupler-isolated driver for quieter, isolated switching.
- Add ambient-light sensing so lights switch on only when needed.
- Add an RTC and data logging for occupancy analytics.
- Design a custom enclosure and move to SMD components for a smaller, production-grade board.
- Add a mains-rated section with proper creepage and slots if switching mains loads directly.

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Appendix

Appendix A: Full Bill of Materials

Ref.	Component	Value / Spec.	Qty	Footprint (KiCad library)	Package
U1	Linear voltage regulator	L7805 (5 V, 1 A)	1	Package_TO_SOT_THT:TO-220-3_Vertical	TO-220
—	ESP32 development board (30-pin)	ESP32 DevKit	1	Plugs into J2 and J3	Module
J2, J3	Female headers for ESP32 (J_LEFT, J_RIGHT)	1×15, 2.54 mm	2	Connector_PinHeader_2.54mm:PinHeader_1x15_P2.54mm_Vertical	THT
J1	DC barrel jack	12 V input	1	Connector_BarrelJack:BarrelJack_Horizontal	THT
J8	USB-C receptacle, power-only 6-pin	USB Type-C	1	Connector_USB:USB_C_Receptacle_HRO_TYPE-C-31-M-12	SMD
SW2	Slide switch SPDT (power switch)	SPDT	1	Button_Switch_THT:SW_Slide_SPDT_Straight_CK_OS102011MS2Q	THT
C1	Electrolytic capacitor (input bulk)	470 µF	1	Capacitor_THT:CP_Radial_Tantal_D8.0mm_P2.50mm	Radial
C3	Electrolytic capacitor (output bulk)	100 µF	1	Capacitor_THT:CP_Radial_Tantal_D8.0mm_P2.50mm	Radial
C2, C4	Ceramic bypass capacitor	100 nF	2	Capacitor_THT:C_Disc_D3.8mm_W2.6mm_P2.50mm	Disc
D2, D4	Schottky diode	1N5819	2	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal	DO-41
D3	Flyback diode (relay coil)	1N4007	1	Diode_THT:D_DO-41_SOD81_P10.16mm_Horizontal	DO-41
D1	Power indicator LED	5 mm, red	1	LED_THT:LED_D5.0mm	5 mm
R1, R4	Resistor (LED / base current limit)	1 kΩ	2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Axial
R2, R3	Pull-up resistor (sensor inputs)	10 kΩ	2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Axial
R5, R6	USB-C CC pull-down resistor	5.1 kΩ	2	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal	Axial
Q1	NPN switching transistor	BC547	1	Package_TO_SOT_THT:TO-92_Inline	TO-92
K1	Relay, SPDT (Form C), 5 V coil	SANYOU SRD series	1	Relay_THT:Relay_SPDT_SANYOU_SRD_Series_Form_C	THT

J7	Screw terminal, 3-pin (relay contacts / load)	5.08 mm pitch	1	TerminalBlock_Phoenix:TerminalBlock_Phoenix_MKDS-1,5-3-5.08_1x03_P5.08mm_Horizontal	THT
J4, J5	IR sensor headers (Sensor_A, Sensor_B)	1×03, 2.54 mm	2	Connector_PinHeader_2.54mm:PinHeader_1x03_P2.54mm_Vertical	THT
J6	OLED I²C header	1×04, 2.54 mm	1	Connector_PinHeader_2.54mm:PinHeader_1x04_P2.54mm_Vertical	THT

Appendix B: Gerber and Drill File List

Files included in the submitted ZIP folder:

- Gerber layers: F_Cu, B_Cu, F_Mask, B_Mask, F_Silkscreen, B_Silkscreen, F_Paste, B_Paste and Edge_Cuts [.gbr]
- Drill files: PTH and NPTH Excellon files [.drl], drill report [.rpt] and job file [.gbrjob]
- Source files: KiCad project (ESP32_Visitor_Counter .kicad_pro, .kicad_sch, .kicad_pcb)
- BOM (.csv) and this documentation (.docx)

Appendix C: Submission Checklist

Deliverable	Included
Project documentation (.docx)	Yes
Gerber and drill files (ZIP)	Yes
Source / design files	Yes
BOM	Yes
Screenshots of schematic, layout, DRC, 3D view and Gerber layers	Yes